

Number and Algebra

Total Marks: 32 marks

Name: _____

Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

**Questions 1-5 are multiple choice one mark each.
Circle the correct answer.**

[5 marks]

1. For the expression $3x^2 - 7x - 3y - 2$. The coefficient of x is

- A** 7 **B** 3 **C** -3
D -7 **E** -2

2. Which of the following pairs are like terms?

- A** $10a$ and $-8a^2$ **B** $6m$ and mn **C** $8st^2$ and $-9s^2t$
D $2t$ and $2e$ **E** $4pq$ and $7pq$

3. Which of the following expressions can be simplified further by combining like terms?

- A** $8 - b + a$ **B** $9a + ab - 5$ **C** $6ab + 4b - ba$
D $7b - ba + 1$ **E** $8a + 2b - 6 + 9ab$

4. Which of the following is equivalent to $3(t + 2)$

- A** $3t$ **B** $3t + 2$ **C** $3t + 6$
D $6t$ **E** $3t + 5$

5. Which of the following is equivalent to $5a + 6 - 3a + 2$

- A** $2a + 4$ **B** $8a + 4$ **C** $10a$
D $2a + 8$ **E** $8a + 8$

6. Simplify the following expressions

a. $2x + 5x$

b. $3 \times a \times b$

c. $\frac{10p}{15}$

d. $7t - 6t - 3$

e. $\frac{9ab}{3b}$

f. $2g \times 6h$

g. $9w - 4w^2 - 6w^2 - 2w$

h. $5cd \times c^2e$

i. $12p^2q \div 15p$

[12 marks]

7. Expand and simplify if needed the following expressions

a. $3(c + 2)$

b. $-5(2c - 4)$

c. $2x(5x - 4)$

d. $3(c + 2) - 5(2c - 4)$

[7 marks]

8. Fully factorise the following Expressions

a. $4p + 12$

b. $10 - 15f$

c. $-2x - 6$

d. $x^2 + 7x$

[8 marks]